Housing Price and Synchronized Admissions between Public and Private Primary Schools: Evidence from Changsha, China*

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Abstract

Using second-hand housing transaction data from Changsha, I perform a simple DID analysis on four top primary schools and their direct neighboring schools to study the effects of the 2020 policy mandating synchronized admission of public and private schools on school district housing price premium. I find that, in contrast to published work that finds increased premium gap, signs of the effects vary largely across schools, and the overall effect of the policy is inconclusive. More in-depth micro analysis is needed to understand how the synchronization policy impacts education equality and the redistribution of education resources across the private and public sectors of education in China.

Keywords: Chinese Education Reform, Equality, Real Estate, School Sector

*This unfinished senior thesis is submitted as a report for ECO4270 Thesis in Economics I. Research limitations in this report and signposts for future research are noted throughout the text using footnotes. I plan to take ECO4280 Thesis in Economics II next semester and finish this project.

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I Introduction

Over the past decade, the net enrollment rate in primary schools among Chinese age-qualified children has increased from 99.85% to over 99.9% (of the People's Republic of China in the Kingdom of Belgium, 2022).[1] In urban areas, education disparity is manifested more in the scarcity of quality education than in the scarcity of education per se. Chinese families compete fiercely to secure seats in top schools throughout each phase of their children's education. Preferences for schools are highly vertical and homogeneous across individuals, i.e., within a given area there is often a shared belief of the ranking of schools with few ties in each rank. State-run public schools remain predominant but private schools have also gained popularity and stature in recent years (Feng, 2023). The landscapes of the comparison or competition between public and private schools vary considerably across cities. In Beijing and Nanjing, public schools provide education with a significantly higher quality than private schools (Chan et al., 2020). However, in Shanghai and Shenzhen, the teaching quality of prestigious private schools matches or exceeds that of high-quality public schools (Jin, Wang and Huang, 2023).

Zeng (2023) documents Chinese public school admission mechanisms in detail. Public primary and middle schools abide by the principle of "nearby enrollment" where registered addresses are the sole criterion of school assignment, and the usage of tests to select students has been strictly forbidden since 2014. Tying school seats to housing has led to staggering school district housing price premiums (Tiebout, 1956; Oates, 1969; Bogart and Cromwell, 2000; Dhar and Ross, 2012; Danielsen, Fairbanks and Zhao, 2015; Zheng, Hu and Wang, 2016; Peng, Tian and Wen, 2021). Numerous methods to balance affordable housing and effective school allocation have been suggested and implemented by economists and policymakers, including promoting neighborhood mobility (DeLuca and Dayton, 2009) and weakening the link between residential location and schooling options through inter-district choice programs (Brunner, Cho and Reback, 2012). Private schools became an important supplement to public education in China around 2000, and they have also been found to significantly improve the spatial supply efficiency of elementary education and reduce education inequality (Wang, 2003; Zhang, Zhao and Wang, 2022). Similar positive effects of private schools on residential de-stratification and public school district premium mitigation are identified by Nechyba (2003), Fack and Grenet (2010), and Sah, Conroy and Narwold (2016).

In 2018, China's Ministry of Education first encouraged a synchronized admission of private and public primary schools nation-wide in its annual guideline notice for basic education enrollment. In 2019, the synchronized admission was mandated with an emphasis on its main objective, namely to eliminate the selection of students by schools in any form for mandatory education. Zeng (2023) studies 134 largest Chinese cities whose population exceeded one million in 2020 and finds that as of 2022, all 114 cities with relevant and retrievable information online have implemented the synchronization policy, but the time of adoption varies largely across cities.[2] Changchun

[1] In China, primary school lasts six years, middle school three years, and high school three years. Primary and middle school education are compulsory. The local Bureau of Education often addresses them using the same policy document. Therefore, this paper jointly discusses primary and middle school education. Secondary education includes middle and high school, which are also referred to as junior and senior secondary school. High schools follow a separate policy document. Basic education refers to twelve years of primary, middle, and high school education.

[2] See Figures 4, 5, and Table A1 in Zeng (2023).

first switched to synchronized admission in 2010, followed by a slow adoption rate of fewer than three new cities per year from 2015 to 2018. 14 cities switched in 2019, and 55 in 2020 (about half of all surveyed cities), as is expected from the national policy. In 2021, the number of new annual adoptions returned to 16 and it has been decreasing thereafter. The scope and coverage of the synchronization policy underlines the significance of research on its implications and in particular, its heterogeneous impacts on different cities.[3] In this paper, I start this quest with using empirical evidence from Changsha, China and a DID model to explore the effect of the policy on school district housing price premium fluctuations. The identification strategy employs a rough combination of the Hedonic Price Model, where house prices are regressed on housing characteristics (Lancaster, 1966; Rosen, 1974) and the school district boundary method, where only houses near the school district boundary are used as samples for regression (Black, 1999).

Before private schools were subject to this restriction, they admitted students according to entrance exams and family backgrounds prior to the public school admission cycle. Consequently, rich families often adopted the dominant strategy of ”try out private schools, fall back on public schools (should the private option fail)” (Huang et al., 2020). This skewed high-quality education resources towards the rich and placed excessive stress on parents maneuvering the system. After the synchronization policy, a penalty in public school lottery priority is incurred on a child who has chosen to enter the lottery for private schools. And if the child’s preferred public school has already fulfilled all seats, the District Education Bureau would intervene and reassign the child to a different school that is often much less desirable.[4] In other words, applying for high-quality private schools would introduce some uncertainty into the quality of public schools that they would fall back upon should they lose the private school lottery. Rich families are then more likely to make a choice between private or public schools, and thereby be restrained from taking more high-quality education opportunities beyond their need. Unifying public and private school admission was intended to promote a more equitable education resource allocation and alleviate parental stress. However, whether this goal has been advanced remains controversial. Since public school allocation depends on the school district where a child’s registered residential address is located, housing prices may serve as one index through which we gauge parental behavioral changes in school selection.

Several papers have executed very similar research designs on Shanghai (Jin, Wang and Huang, 2023), Hangzhou (Tian, Teng and Ji, 2023), and Chengdu (Chen and Li, 2023). All three papers conclude that the policy results in a significant appreciation in school district housing prices, with a stronger effect on more elite and desirable school districts. The synchronization policy potentially exacerbates education inequality to the contrary of its intentions. However, these papers share the following limitations. First, the housing characteristics that are used as controls in the DID regression are all given by the Lianjia website, and almost all given non-price variables are controlled for. Thus there is non-negligible arbitrariness in the selection of controls, and this selection process is not based

[3]Originally, I planned to take advantage of having surveyed 134 cities and study this city-heterogeneous effect. Changsha ended up to be what I started with, but I still look forward to extending analysis to across-city comparisons. No prior research on this policy has gone beyond one specific city.

[4]Cities vary in how the opportunity cost is designed and come cities’ design does not incur such an opportunity cost at all. For more information, please see the dataset documented in Zeng (2023).

on economic theory and arguments, but rather on availability of data from the same provider. Second, these papers do not present a solid basis for the usage of Lianjia's data only. Despite its strong presence in the domestic real estate market, its market share and market coverage vary largely across cities. For example, in Beijing, Lianjia and Wo-Ai-Wo-Jia form a duopoly in the second-hand real estate agency. They do not split the market by house price, characteristics, or location, so the Lianjia sample is likely a representative sample of the market (Zhong, 2018). However, in Chengdu, Shanghai, and Hangzhou, Lianjia entered the market much later compared to Beijing and faced fiercer competition from more companies. Its data coverage fluctuated by year from 20% to 50% without explicit evidence against correlation with price, housing characteristics and location.[5] Third, a complete sorting of all schools into different tiers and estimating each tier's housing price premium is an inaccurate representation of parental psychology. Parents' shared belief on school ranking should be our primary basis for school sorting. At least in Changsha, parents' perception of schools is more akin to a dichotomous top or non-top school rather than dividing schools into different tiers.[6] Fourth, despite a unanimous conclusion that the synchronization policy may harm education equality, the specific mechanism through which this causal chain is complete is either lacking or unconvincing.

This paper shares some of the same issues that are mentioned above. But it provides a critique to an early consensus in the evaluation of the synchronized admission policy. Using Lianjia second-hand housing transaction price data in Changsha and zooming in on four top-10 primary schools and their neighboring non-top 10 primary schools, I show that the policy effect is very likely inconclusive on the city level due to large heterogeneous on the school level.

II Data

The housing data in this paper is obtained from Lianjia, the largest real estate intermediary company in China. The record of detailed property trading information, including final transaction prices, transaction dates, architectural characteristics, community characteristics, and neighbourhood location characteristics, is publicly available on Lianjia 's webpage[7]. 50,077 secondary housing transaction records for six urban districts in Changsha are collected from 2016 to 2023.

School district data are accessed and scraped from an official local newspaper portal.[8]

[5] My paper using Lianjia to study Changsha faces the same problem. To address this problem, I may need to incorporate data from multiple second-hand real estate agencies.

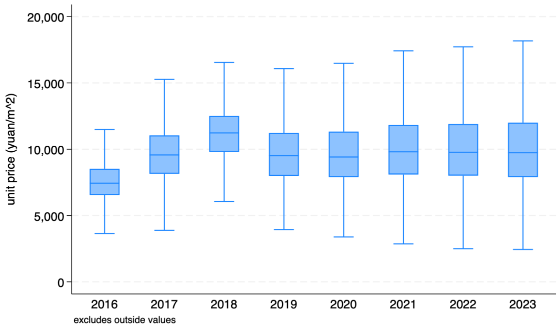
[6] However, my paper only examines primary school ranking due to data limitations. To be more realistic, I should examine at least primary and middle school in the vicinity because parents rarely only consider primary school opportunities. It is very likely that they put more weight on middle school district because it is directly relevant to the high school entrance exam.

[7] https://www.lianjia.com

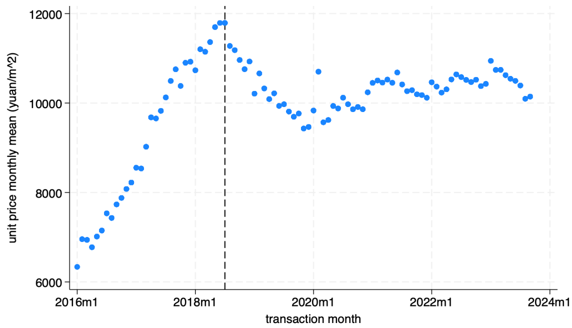
[8] The newspaper is called "Changsha Wan Bao," and it provides a portal that would return the corresponding primary school or real estate names if either one of the pair entered into the search box. Here are the websites for 2018, 2019, 2020, 2021, 2022, 2023. I impute the information for 2016 and 2017 using that of 2018 since this website only started in 2018. I use selenium to prompt several keywords that collectively return all entries of primary schools in the database, scrape school district data by year, and later match them with school names.

Figure 1: Mean Unit Housing Price Over Time

(a) Box Plot of Mean Unit Price by Transaction Year



(b) Scatter Plot of Mean Unit Price by Transaction Month

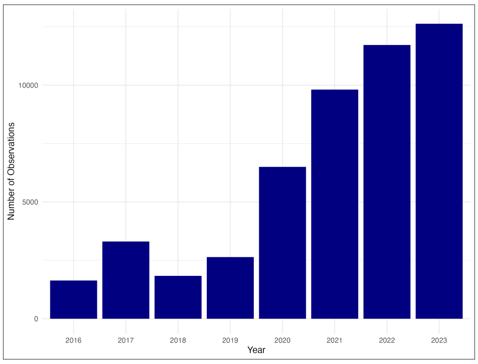


Notes: These figures show the evolution of second-hand housing transaction unit price over time in Changsha.

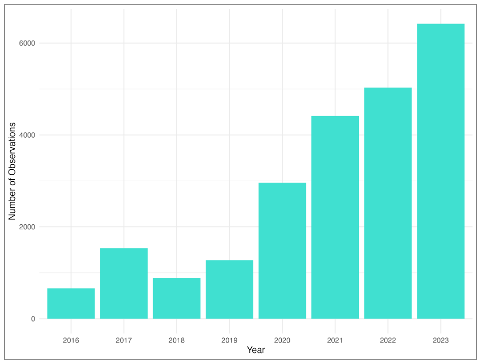
Figure 1 plots the mean unit transaction price of second-hand houses in Changsha by year and by month.[9]

Figure 2: Numbers of Observations by Year and District

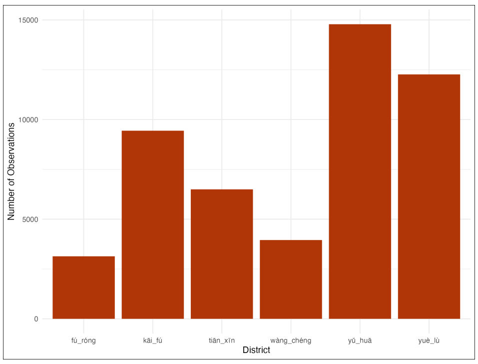
(a) By Year (Full, 50077 obs)



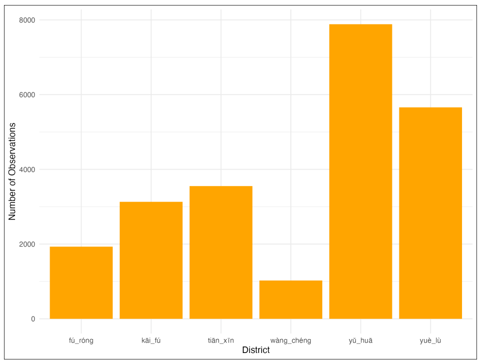
(b) By Year (Matched, 23182 obs)



(c) By District (Full, 50077 obs)



(d) By District (Matched, 23182 obs)



Notes: These figures plot the distribution of observations before and after matching Lianjia data with school district data.

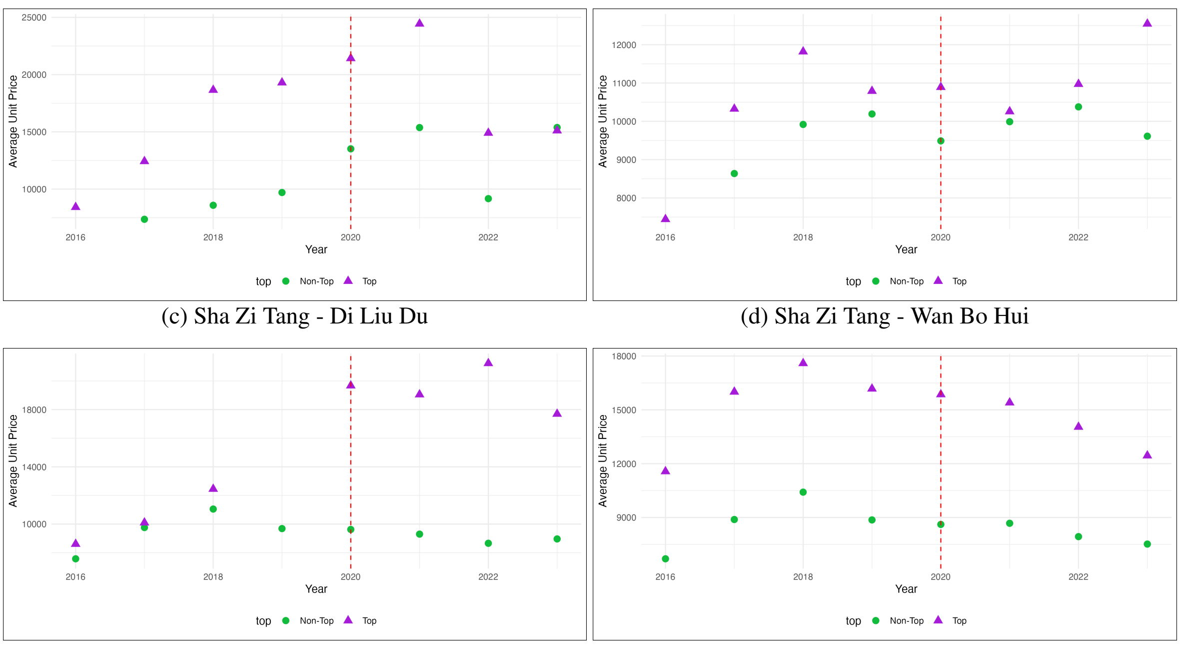
From Figure 2 we can see that after matching Lianjia data with school district data, the distribution of observations across and year and district remain largely similar. However, the original full sample Lianjia data is clearly

[9] I need to look into why the trend looks like this, but it is not that relevant to this paper since I do not look at price premiums itself but its changes. Do I need to adjust price by inflation?

not proportionately balanced across year and district. This is possibly due to the market status of Lianjia relative to other real estate brokerage firms in Changsha.

Aggregating information from education counselling blogs, school district housing websites, school selection service websites and parent communities, 13 primary schools are categorized as ”top” schools. I was not able to derive from the internet a more granular partition of primary schools into more than two tiers (top and non-top). However, after another round of matching, only fewer than 500 out of the 20 thousand transaction observations are matched with a top school.[10] To ensure the quality of the counterfactual, the sample of interest is narrowed down to only the top primary schools and their directly adjacent schools.[11] The observations of 9 out of the 13 top schools are either too few (1 to 5 observations per year) or ill-structured (almost all observations occur after 2020) to perform a DID with a cutoff at 2020. The four top schools that eligible for analysis are Sha Zi Tang Elementary School - Di Liu Du (SZT-DLD), Sha Zi Tang Elementary School - Wan Bo Hui (SZT-WBH), Qing Yuan Elementary School (QY), and Lu Shan International Experimental Elementary School (LS).[12]

Figure 3: Unit Price by Year and Elementary School



(a) Lu Shan International Experimental

(b) Qing Yuan

(c) Sha Zi Tang - Di Liu Du

(d) Sha Zi Tang - Wan Bo Hui

Notes: These figures plot the unit price change from 2016 to 2023 for top schools and their corresponding adjacent non-top schools.

[10]One of the reasons for many failures in matching is that the elementary school names and the real estate names from the two different data sources are both text entries with many inherent errors, and that the same real estate can have slightly different names on the two platforms, making a direct match very unsatisfactory. Later for the four selected top primary schools, I manually checked each elementary school entry for each year and corrected them to match the real estate names provided by Lianjia. This is extremely time-consuming and it does not feel realistic to perform this on a larger scale. I am not sure whether the three studies ignored this problem or used more advanced matching methods like LLM or directly geocoded boundary matching.

[11]This information is retrieved from this website. See Appendix section for an example of the webpage.

[12]I did not have time to finish formatting and interpreting the balance tables for each school and their adjacent counterfactuals. Therefore I put them in the appendix. I think I want to make the table to be by year and by variable ”top,” but I still need to figure out how to do that. The thing is, even if they do not satisfy the qualifications of a good counterfactual, I feel like there’s nothing much I can do about it with the limited data I have? I also did not have time to rigorously test the parallel trend assumption.

From Figure 3 we can see that the top elementary school housing price premium does not exhibit a significant change before and after 2020 for LS and STZ-WBH, but the effect for QY seems to be negative, while the effect for SZT-DLD seems to be significantly positive.

III Empirical Strategy

To assess the impact of the synchronous enrollment policy on school district housing prices, I begin with a simple DID model.

$$ln(UnitPrice_{it}) = \alpha + \beta Top_i \times Post_t + \delta Top_i + \theta Post_t + \gamma X_{it} + \varepsilon_{it}$$

where i is the individual real estate, t is year, $ln(UnitPrice_{it})$ is the natural log of the unit price of real estate i at time t, Top_i is a dummy variable that takes on value 1 if the real estate i is located in a school district with a top school and 0 otherwise, $Post_t$ is a dummy variable that takes the value of 1 after 2020 and 0 otherwise, X_{it} is a vector of control variables comprised of housing characteristics (including interior finishing type, layout/floor plan, floor, construction type, whether the building has an elevator, construction structure, and number of subways within 2km[13]), $\varepsilon_i t$ is the error term. I do not incorporate year fixed effects and district fixed effects because the main four regressions are on the micro school level and these fixed effects are not necessary.[14] β is our DID estimator of interest.

IV Results

Table 1 presents the DID regression results for the four top schools both separately and pooled together. At 5 percent significance level, the correlation between the implementation of the synchronization policy and top school district premium is significantly positive for LS and SZT-DLD, and significantly negative for SZT-WBH. Pooled together, the correlation between the policy and housing premiums becomes insignificant.

The synchronization policy very likely affected each public school in very different ways, and the factors at play are too complicated to be captured in DID models that encompass almost all schools in a city.[15] For example, a unique feature of QY is that it is geographically surrounded by its affiliated campuses, which presumably are not as high-quality as the main campus but are considered to be much better than the direct neighbors of most top schools. SZT-DLD's assigned real estates were only gradually built around 2016-2020, this may also have affected price changes. Given these considerable heterogeneity, it is very hard to make a conclusion about the overall effect of the policy.

[13] These are all the housing characteristics that are available to me in the data I have. But as I mentioned in Section I, choosing controls solely according to availability of data is arbitrary not rigorous enough.

[14] I think it makes some sense to add school fixed effects in the pooled DID model, but this also feels pointless since these four top schools are not proven to be representative of all top schools - they are very likely not representative.

[15] I thought of some very intuitive reasons for why the effects are so different for these four schools based on my knowledge of their respective areas, but I didn't have enough time to verify my guesses.

Table 1: DID Regression Results

	Dependent variable:				
	log(UnitPrice)				
	(1)	(2)	(3)	(4)	(5)
	LS	QY	SZT-DLD	SZT-WBH	All
Top	0.094	0.052^{**}	0.114^{***}	0.445^{***}	0.300^{***}
	(0.085)	(0.025)	(0.029)	(0.025)	(0.016)
Post	-0.117^{*}	-0.014	-0.087^{***}	-0.057^{**}	-0.009
	(0.070)	(0.020)	(0.012)	(0.023)	(0.012)
Top×Post	0.493^{***}	0.054^{*}	0.636^{***}	-0.067^{**}	-0.030
	(0.103)	(0.029)	(0.046)	(0.032)	(0.021)
Constant	8.349^{***}	8.954^{***}	9.017^{***}	9.000^{***}	9.103^{***}
	(0.405)	(0.115)	(0.168)	(0.192)	(0.115)
Obs	476	723	777	989	2,600
R^2	0.449	0.442	0.565	0.661	0.452
Adjusted R^2	0.408	0.405	0.541	0.642	0.438
RSE	0.371 (df=442)	0.164 (df=677)	0.158 (df=735)	0.233 (df=937)	0.244 (df=2534)
F Statistic	10.915^{***}	11.935^{***}	23.296^{***}	35.757^{***}	32.123^{***}
	(df = 45; 677)	(df = 45; 677)	(df = 41; 735)	(df = 51; 937)	(df = 65; 2534)
Note:					*p<0.1; **p<0.05; ***p<0.01

V Conclusion

In this paper, I use a simple DID model on four top primary schools and their direct neighboring schools in Changsha to gauge the effects of the 2020 policy mandating public and private schools synchronize admission on school district housing price premium. I find that the signs of the effects vary largely across schools, and the overall effect of the policy after pooling the four samples together are insignificant. This study performs a DID analysis very similar to that of Jin, Wang and Huang (2023), Tian, Teng and Ji (2023), and Chen and Li (2023) but on a much more micro unit, with specific top schools as the treatment group and their directly adjacent non-top schools as the control group. The findings suggest that a DID analysis using almost all schools in a city as the sample may produce biased results due to ill-sampling (over-relying on Lianjia as the source of second-hand housing transaction data irrespective of its dynamic market share in a given city), and it may also overlook huge heterogeneity between schools and falsely arrive at positive or negative conclusions about the policy effect when the aggregate effect is in fact inconclusive.

VI Appendix

Table 2: Lu Shan International Experimental Elementary School

Char	**N**	**Overall**, N = 476	**2016**, N = 4	**2017**, N = 27	**2018**, N = 15	**2019**, N = 22	**2020**, N = 59	**2021**, N = 51	**2022**, N = 75	**2023**, N = 223
tranpri	476	129 (88, 213)	79 (75, 87)	106 (90, 127)	148 (141, 225)	156 (49, 200)	185 (159, 240)	227 (118, 266)	99 (71, 119)	130 (89, 222)
unitpri	476	13,032 (8,762, 20,459)	8,336 (7,995, 8,757)	11,539 (10,210, 14,160)	18,206 (14,195, 19,584)	17,991 (8,742, 20,252)	19,496 (13,103, 21,755)	23,962 (10,787, 26,859)	8,919 (7,521, 10,886)	13,636 (9,352, 21,642)
top	476									
Non-Top		355 (75%)	0 (0%)	1 (3.7%)	3 (20%)	10 (45%)	30 (51%)	25 (49%)	74 (99%)	212 (95%)
Top		121 (25%)	4 (100%)	26 (96%)	12 (80%)	12 (55%)	29 (49%)	26 (51%)	1 (1.3%)	11 (4.9%)
after_2020	476	349 (73%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	51 (100%)	75 (100%)	223 (100%)
floor	476									
Middle		180 (38%)	1 (25%)	7 (26%)	7 (47%)	4 (18%)	24 (41%)	20 (39%)	28 (37%)	89 (40%)
Low		140 (29%)	2 (50%)	12 (44%)	3 (20%)	5 (23%)	17 (29%)	13 (25%)	18 (24%)	70 (31%)
High		156 (33%)	1 (25%)	8 (30%)	5 (33%)	13 (59%)	18 (31%)	18 (35%)	29 (39%)	64 (29%)
constrtype	476									
Ta		49 (10%)	0 (0%)	0 (0%)	2 (13%)	5 (23%)	3 (5.1%)	7 (14%)	11 (15%)	21 (9.4%)
NA		22 (4.6%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	17 (23%)	5 (2.2%)
Ban+TA		268 (56%)	4 (100%)	27 (100%)	12 (80%)	13 (59%)	31 (53%)	28 (55%)	26 (35%)	127 (57%)
Ban		137 (29%)	0 (0%)	0 (0%)	1 (6.7%)	4 (18%)	25 (42%)	16 (31%)	21 (28%)	70 (31%)
elev	476									
No		7 (1.5%)	0 (0%)	0 (0%)	1 (6.7%)	2 (9.1%)	1 (1.7%)	2 (3.9%)	0 (0%)	1 (0.4%)
NA		23 (4.8%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	1 (2.0%)	8 (11%)	14 (6.3%)
Yes		446 (94%)	4 (100%)	27 (100%)	14 (93%)	20 (91%)	58 (98%)	48 (94%)	67 (89%)	208 (93%)

Table 3: Qing Yuan Elementary School

Char	**N**	**Overall**, N = 794	**2016**, N = 5	**2017**, N = 49	**2018**, N = 36	**2019**, N = 70	**2020**, N = 80	**2021**, N = 144	**2022**, N = 159	**2023**, N = 251
tranpri	794	113 (82, 148)	82 (68, 100)	93 (74, 109)	117 (80, 159)	115 (76, 148)	93 (68, 136)	111 (80, 136)	115 (88, 161)	124 (87, 155)
unitpri	794	10,143 (8,884, 11,638)	7,875 (6,579, 8,128)	9,410 (8,260, 10,882)	11,469 (10,353, 12,273)	10,845 (9,560, 11,552)	10,120 (8,923, 10,898)	9,965 (8,909, 11,134)	10,108 (9,065, 11,928)	10,230 (8,633, 12,406)
top	794									
Non-Top		533 (67%)	0 (0%)	24 (49%)	12 (33%)	38 (54%)	57 (71%)	111 (77%)	123 (77%)	168 (67%)
Top		261 (33%)	5 (100%)	25 (51%)	24 (67%)	32 (46%)	23 (29%)	33 (23%)	36 (23%)	83 (33%)
after_2020	794	554 (70%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	144 (100%)	159 (100%)	251 (100%)
floor	794									
Middle		325 (41%)	1 (20%)	19 (39%)	10 (28%)	27 (39%)	30 (38%)	61 (42%)	72 (45%)	105 (42%)
Low		198 (25%)	2 (40%)	16 (33%)	12 (33%)	17 (24%)	22 (28%)	40 (28%)	34 (21%)	55 (22%)
High		271 (34%)	2 (40%)	14 (29%)	14 (39%)	26 (37%)	28 (35%)	43 (30%)	53 (33%)	91 (36%)
constrtype	794									
Ta		66 (8.3%)	0 (0%)	6 (12%)	2 (5.6%)	6 (8.6%)	12 (15%)	15 (10%)	9 (5.7%)	16 (6.4%)
Ping		4 (0.5%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	3 (2.1%)	0 (0%)	1 (0.4%)
NA		8 (1.0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	1 (1.3%)	1 (0.7%)	2 (1.3%)	4 (1.6%)
Ban+TA		504 (63%)	5 (100%)	28 (57%)	28 (78%)	50 (71%)	37 (46%)	83 (58%)	106 (67%)	167 (67%)
Ban		212 (27%)	0 (0%)	15 (31%)	6 (17%)	14 (20%)	30 (38%)	42 (29%)	42 (26%)	63 (25%)
elev	794									
No		48 (6.0%)	0 (0%)	0 (0%)	0 (0%)	5 (7.1%)	5 (6.3%)	8 (5.6%)	11 (6.9%)	19 (7.6%)
NA		32 (4.0%)	0 (0%)	0 (0%)	0 (0%)	1 (1.4%)	3 (3.8%)	8 (5.6%)	7 (4.4%)	13 (5.2%)
Yes		714 (90%)	5 (100%)	49 (100%)	36 (100%)	64 (91%)	72 (90%)	128 (89%)	141 (89%)	219 (87%)

Table 4: Sha Zi Tang Elementary School - Di Liu Du

Char	**N**	**Overall**, N = 777	**2016**, N = 36	**2017**, N = 103	**2018**, N = 82	**2019**, N = 79	**2020**, N = 140	**2021**, N = 99	**2022**, N = 121	**2023**, N = 117
tranpri	777	78 (53, 111)	84 (55, 116)	70 (51, 99)	93 (57, 123)	79 (52, 105)	79 (43, 110)	74 (50, 116)	73 (55, 108)	86 (63, 121)
unitpri	777	9,578 (8,325, 10,896)	7,940 (6,781, 8,984)	9,767 (8,895, 10,654)	11,313 (10,712, 12,109)	9,807 (8,483, 11,011)	9,668 (8,953, 10,456)	9,506 (8,312, 10,438)	8,584 (7,262, 9,966)	8,920 (8,086, 11,239)
top	777									
Non-Top		698 (90%)	22 (61%)	82 (80%)	68 (83%)	79 (100%)	138 (99%)	92 (93%)	115 (95%)	102 (87%)
Top		79 (10%)	14 (39%)	21 (20%)	14 (17%)	0 (0%)	2 (1.4%)	7 (7.1%)	6 (5.0%)	15 (13%)
after_2020	777	337 (43%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	99 (100%)	121 (100%)	117 (100%)
floor	777									
Middle		339 (44%)	13 (36%)	41 (40%)	35 (43%)	40 (51%)	59 (42%)	44 (44%)	60 (50%)	47 (40%)
Low		177 (23%)	9 (25%)	23 (22%)	21 (26%)	14 (18%)	35 (25%)	27 (27%)	23 (19%)	25 (21%)
High		261 (34%)	14 (39%)	39 (38%)	26 (32%)	25 (32%)	46 (33%)	28 (28%)	38 (31%)	45 (38%)
constrtype	777									
Ta		131 (17%)	4 (11%)	23 (22%)	13 (16%)	7 (8.9%)	20 (14%)	15 (15%)	19 (16%)	30 (26%)
NA		1 (0.1%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	1 (0.8%)	0 (0%)
Ban+Ta		388 (50%)	29 (81%)	54 (52%)	51 (62%)	39 (49%)	76 (54%)	43 (43%)	46 (38%)	50 (43%)
Ban		257 (33%)	3 (8.3%)	26 (25%)	18 (22%)	33 (42%)	44 (31%)	41 (41%)	55 (45%)	37 (32%)
elev	777									
No		111 (14%)	1 (2.8%)	4 (3.9%)	9 (11%)	16 (20%)	22 (16%)	20 (20%)	26 (21%)	13 (11%)
NA		3 (0.4%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	1 (0.7%)	0 (0%)	2 (1.7%)	0 (0%)
Yes		663 (85%)	35 (97%)	99 (96%)	73 (89%)	63 (80%)	117 (84%)	79 (80%)	93 (77%)	104 (89%)

Table 5: Sha Zi Tang Elementary School - Wan Bo Hui

Char	**N**	**Overall**, N = 989	**2016**, N = 98	**2017**, N = 94	**2018**, N = 59	**2019**, N = 86	**2020**, N = 178	**2021**, N = 164	**2022**, N = 169	**2023**, N = 141
tranpri	989	83 (56, 121)	70 (56, 104)	90 (73, 132)	99 (85, 145)	82 (51, 108)	82 (51, 120)	91 (57, 132)	81 (57, 132)	73 (48, 110)
unitpri	989	10,423 (7,895, 14,804)	10,607 (8,595, 12,008)	15,033 (9,799, 17,334)	15,458 (10,885, 19,595)	11,329 (8,789, 18,628)	10,022 (7,931, 16,411)	9,871 (7,939, 14,392)	8,725 (7,222, 13,209)	8,831 (6,823, 12,138)
top	989									
Non-Top		480 (49%)	24 (24%)	28 (30%)	20 (34%)	34 (40%)	98 (55%)	99 (60%)	100 (59%)	77 (55%)
Top		509 (51%)	74 (76%)	66 (70%)	39 (66%)	52 (60%)	80 (45%)	65 (40%)	69 (41%)	64 (45%)
after_2020	989	474 (48%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	164 (100%)	169 (100%)	141 (100%)
floor	989									
Middle		441 (45%)	37 (38%)	45 (48%)	24 (41%)	39 (45%)	85 (48%)	63 (38%)	86 (51%)	62 (44%)
Low		249 (25%)	33 (34%)	16 (17%)	20 (34%)	23 (27%)	39 (22%)	40 (24%)	34 (20%)	44 (31%)
Basem		2 (0.2%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	2 (1.2%)	0 (0%)	0 (0%)
High		297 (30%)	28 (29%)	33 (35%)	15 (25%)	24 (28%)	54 (30%)	59 (36%)	49 (29%)	35 (25%)
constrtype	989									
Ta		137 (14%)	6 (6.1%)	16 (17%)	8 (14%)	13 (15%)	23 (13%)	24 (15%)	25 (15%)	22 (16%)
Ping		4 (0.4%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	3 (1.8%)	1 (0.7%)
NA		5 (0.5%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	2 (1.2%)	3 (2.1%)
Ban+Ta		542 (55%)	69 (70%)	63 (67%)	42 (71%)	44 (51%)	82 (46%)	82 (50%)	85 (50%)	75 (53%)
Ban		301 (30%)	23 (23%)	15 (16%)	9 (15%)	29 (34%)	73 (41%)	58 (35%)	54 (32%)	40 (28%)
elev	989									
No		198 (20%)	3 (3.1%)	3 (3.2%)	4 (6.8%)	17 (20%)	50 (28%)	43 (26%)	45 (27%)	33 (23%)
NA		24 (2.4%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	4 (2.2%)	4 (2.4%)	8 (4.7%)	8 (5.7%)
Yes		767 (78%)	95 (97%)	91 (97%)	55 (93%)	69 (80%)	124 (70%)	117 (71%)	116 (69%)	100 (71%)

Table 6: All Four Schools

Char	**N**	**Overall**, N = 3,036	**2016**, N = 143	**2017**, N = 273	**2018**, N = 192	**2019**, N = 257	**2020**, N = 457	**2021**, N = 458	**2022**, N = 524	**2023**, N = 732
tranpri	3,036	96 (63, 138)	74 (56, 104)	89 (62, 118)	100 (70, 147)	89 (57, 133)	90 (53, 137)	100 (66, 142)	95 (66, 132)	111 (71, 160)
unitpri	3,036	10,149 (8,423, 12,657)	9,325 (7,402, 11,460)	10,350 (9,009, 13,573)	11,616 (10,683, 13,520)	10,477 (8,864, 12,272)	10,003 (8,496, 12,704)	9,999 (8,419, 12,446)	9,380 (7,722, 11,679)	10,315 (8,227, 13,772)
top	3,036									
Non-Top		2,066 (68%)	46 (32%)	135 (49%)	103 (54%)	161 (63%)	323 (71%)	327 (71%)	412 (79%)	559 (76%)
Top		970 (32%)	97 (68%)	138 (51%)	89 (46%)	96 (37%)	134 (29%)	131 (29%)	112 (21%)	173 (24%)
after_2020	3,036	1,714 (56%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	458 (100%)	524 (100%)	732 (100%)
floor	3,036									
Middle		1,285 (42%)	52 (36%)	112 (41%)	76 (40%)	110 (43%)	198 (43%)	188 (41%)	246 (47%)	303 (41%)
Low		764 (25%)	46 (32%)	67 (25%)	56 (29%)	59 (23%)	113 (25%)	120 (26%)	109 (21%)	194 (27%)
Basem		2 (¡0.1%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	2 (0.4%)	0 (0%)	0 (0%)
High		985 (32%)	45 (31%)	94 (34%)	60 (31%)	88 (34%)	146 (32%)	148 (32%)	169 (32%)	235 (32%)
constrtype	3,036									
Ta		383 (13%)	10 (7.0%)	45 (16%)	25 (13%)	31 (12%)	58 (13%)	61 (13%)	64 (12%)	89 (12%)
Ping		8 (0.3%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	3 (0.7%)	3 (0.6%)	2 (0.3%)
NA		36 (1.2%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	1 (0.2%)	1 (0.2%)	22 (4.2%)	12 (1.6%)
Ban+Ta		1,702 (56%)	107 (75%)	172 (63%)	133 (69%)	146 (57%)	226 (49%)	236 (52%)	263 (50%)	419 (57%)
Ban		907 (30%)	26 (18%)	56 (21%)	34 (18%)	80 (31%)	172 (38%)	157 (34%)	172 (33%)	210 (29%)
elev	3,036									
No		364 (12%)	4 (2.8%)	7 (2.6%)	14 (7.3%)	40 (16%)	78 (17%)	73 (16%)	82 (16%)	66 (9.0%)
NA		82 (2.7%)	0 (0%)	0 (0%)	0 (0%)	1 (0.4%)	8 (1.8%)	13 (2.8%)	25 (4.8%)	35 (4.8%)
Yes		2,590 (85%)	139 (97%)	266 (97%)	178 (93%)	216 (84%)	371 (81%)	372 (81%)	417 (80%)	631 (86%)

Figure 4: Example of the Adjacent-District Finder Webpage

Notes: This is a snapshot from webpage https://hunan.tianditu.gov.cn/TDTHN/portal/xuequ2023/index.html/.

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